Insert Your Capstone Article Title Here

Insert Subtitle Here (if applicable)

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GENERAL WRITING GUIDELINES

DELETE THIS SECTION BEFORE FINAL SUBMISSION

This report should be written in a formal style, similar to an academic article or a professional white paper. You should write for an audience of semi-technical stakeholders, such as a VP of engineering or a CTO. You can assume some technical knowledge, but keep the details at a level that would be interesting to this group. For example, in describing your model architecture, you should give details about the number of hidden layers, nodes, and activation functions used, but not the specific lines of code or names of function calls used to build it.

Under each section of this template, and in the weekly assignment instructions in Canvas, you will find a list of questions that are meant to guide you in terms of what information should be included in each section. DO NOT SUBMIT A LIST OF QUESTIONS AND ANSWERS. A reader should be able to answer these questions after reading your text if they are relevant to your project; they do not need to be answered explicitly, and you should avoid repeating information in your report just for the sake of answering each question independently.

A selection of formatting examples can be found in Section 7 of this document. Please refer to these examples for the proper way to include figures, tables, equations, and in-text citations, along with other general formatting guidelines.

ABSTRACT

The abstract should be the last piece of your report that you write and will be a 1-2 paragraph overview of your entire project. As a rough guide, you should expect to have 1-2 sentences introducing the problem and your chosen dataset, 3-4 sentences describing your chosen methods, and 1-2 sentences summarizing your project results.

KEYWORDS

Insert Applicable Keywords Here. Insert Applicable Keywords Here.

1 Introduction

This section should be used to introduce your project at the highest level. The reader should be able to answer the following questions:

- What question are you answering or what problem are you solving with your project?
- Why is this project important or useful?
- Who is the end user of your AI product/model?
- What data will you be using in your project? Where does this data come from? If your project is meant to simulate a live AI product, where would the data come from in the live system?
- What is the ultimate goal of your project? If you are answering a question, what is your (high level) hypothesis? If you are solving a problem, what would the final product look like?

2 Data Summary

This section of your final report should include details about your dataset and your exploratory data analysis. The reader should be able to answer the following questions:

- What variables are present in your dataset? Describe these in plain language rather than using the variable names from your code.
- What issues were present in your dataset and what steps were taken to handle them (missing data, regularization, etc.)? Do you know the source of these data issues, and how would you deal with these issues in a fully developed machine learning pipeline?
- How are your variables related to your project goal? Do you see any patterns in the data that would suggest that they are/are not going to be useful in your machine learning model(s)?
- How are your variables related to each other? Are there any strong correlations? How might this affect how you build your machine learning model(s)?

3 Literature Review

This section of your final report should give background information about your project application/research question and the machine learning method(s) that you have chosen to use. This section should be completely focused on existing work, rather than on your own methods and choices. For example, instead of writing "*We chose method x for application y because it has benefit z*", you would write "*(Citation) developed method x for application y and found that it had benefit z*". Then, you can reference this shown benefit when you introduce your own methods in the next section.

The reader should be able to answer the following questions:

- What other methods/technologies exist that attempt to solve the same problem/answer the same question that your project will address? This can include academic research or commercial projects.
- How was your machine learning method/system developed and what are its benefits (especially those relevant to your project)?
- What existing work demonstrates that these method(s) are appropriate (or particularly well-suited) to use in your project?
- What are some examples of similar projects that use the method(s) that you have chosen? Again, you can include either academic research or commercial projects.

4 Methodology

This section of your final report should describe the final model architecture or pipeline choices that you have made, as well as the training and optimization procedures that you follow. Training results should be saved for the Results section of the paper—any discussion of modeling iterations and optimization should be brief, without any mention of the resulting performance. For example: "*During model optimization, we explored hyperparameters a, b, and c with values [d,e,f], [g,h,i], and [j,k,l], respectively. Our final model architecture was...*" If the comparative results over your hyperparameter search are interesting enough to be included, these should be discussed in the results section.

The reader should be able to answer the following questions after reading your methodology:

- What type of machine learning model(s) are you using?
- What are the notable architectural or system design choices that you have made and why? E.g., number of nodes, hidden layers, activation functions, etc.
- What is your model training procedure or machine learning pipeline? E.g., How did you split your data? How many epochs/iterations did you use during training? What was your batch size? What loss function/training metrics were used?
- How did you optimize your model? E.g., Which hyperparameters and/or architectural elements did you adjust?

5 Results

This section of your final report should describe how your model/system performed and provide evidence for how your work did or did not support your experimental hypothesis or solve your chosen problem. The reader should be able to answer the following questions:

- How effective was your machine learning model at learning the task? e.g. Did the model overfit/underfit the training data? What does the training accuracy/loss curve look like for your model?
- If your project is focused on answering a research question, what evidence do you have to support or disprove your hypothesis?
- If your project is focused on solving a problem, how will your model performance affect the application of your model to that problem?
- If your project is focused on implementing a complete machine learning system, what is the user experience? Are there any delays or remaining bugs that affect your model output either in common or edge use cases?

6 Conclusion

This section of your final report should provide a very brief summary of your entire project. It should highlight the most interesting or significant results and discuss any future work that you or another researcher might do. The reader should be able to answer the following questions:

- What was the main hypothesis or goal of the project, and how did your models perform in relation to this hypothesis/goal?
- What was the most interesting or significant result of your project?
- Did your model training or performance bring about any unexpected results?
- If you were to continue this project, what would your next steps be? For example,

would you continue to optimize your model or try other model architectures? Would you change or expand the data used in your model? What steps need to be taken to "productionize" your model?

7 FORMATTING EXAMPLES

DELETE THIS SECTION BEFORE FINAL SUBMISSION

7.1 General Formatting

A formal report of this type should be primarily in paragraph form, with minimal subheadings, bullet points, or numbered lists. For most of your required report sections, no subheadings will be necessary, and the information can be presented as a single, cohesive discussion of the content relevant to that section. As a general rule of thumb, bulleted or numbered lists are only appropriate in this kind of report if the list items are less than one sentence, but are too long/detailed to be presented in a table.

If you feel strongly that your content would benefit from subheadings or bulleted/numbered lists, they should follow the formatting examples used in this section.

7.2 In-text Citations

A report of this type should be heavily supported by in-text citations. You should include a new supporting citation in each section for:

1. The first time that you mention a specific code library or dataset,
2. The first time that you introduce a pre-trained model or model architecture,
3. Any information that is not your own original thought/idea, even if you didn't need to look it up.

When in doubt, it is always better to over-cite than to under-cite.

Formatting rules for in-text citations can be found in the APA guidelines, but examples of the three most common uses are discussed in the following paragraphs.

Parenthetical citations at the end of a sentence should be used when the information from the entire sentence comes from the cited source (American Psychological Association, 2025).

Parenthetical citations mid-sentence should be used when only a part of the sentence comes from the cited source, or when you are providing a citation for something specific such as a company, model architecture/method, or technique. See the following examples for a sentence with multiple mid-sentence citations and a sentence with a specific mid-sentence citation below:

- The transformer model was introduced in 2017 for the application of machine translation (Vaswani et al., 2017) and quickly became the most common backbone for today's large language models (Devlin et al., 2019; Radford et al., 2019).
- As one of the first transformer-based generative text models, OpenAI's GPT (Radford et al., 2019) quickly became the standard large language architecture used for generative tasks.

Narrative in-text citations can be used when you want to include the name of the author(s) in your main text, as shown in the following example:

> Vaswani et al. (2017) originally developed the transformer architecture for the application of machine translation.

7.3 Figures, Tables, and Equations

Most of your supporting content should be included in the main text of the report, rather than at the end. While the in-line formatting can be a headache, this makes it easier for your reader to reference your figures and tables while reading, without having to scroll back and forth too much. See Section 7.5 for more details about how to properly use appendices for supplemental information.

7.3.1 Figures

Figures should be numbered sequentially *below* the image and referred to directly in your text (e.g. "As seen in Figure 7.1 [...]", "Figure 7.1 shows [...]"). If your figure is important enough to include in your report, then it deserves at least one sentence of discussion in the text.

Each figure should also have a corresponding caption that briefly describes the content of the figure and can be used to highlight important details. If your figure is too complex to be limited to a single column width, you can include it as a full-width image at the top of the page.

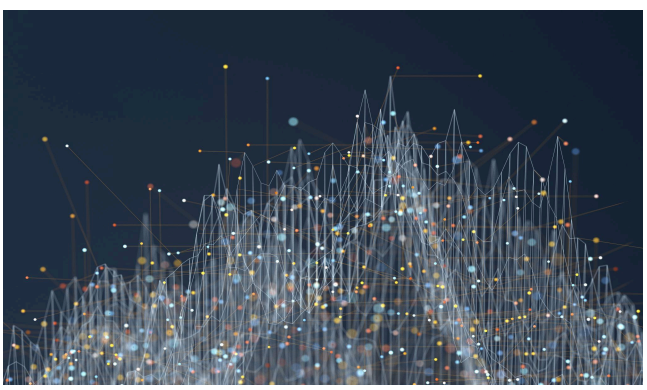

Figure 7.1

Example text: Model accuracy for our six models during training. The best performing model is shown in red.

7.3.2 Tables

Tables should be numbered sequentially above the table and should have an in-text mention and a descriptive caption, like the figure examples given above.

Feel free to improve the default MS Word table formatting shown here for aesthetic appeal (e.g. change cell size, border width, font size, etc.). If you have a favorite table formatting tool (e.g. LaTeX, Pandas pretty print, etc.), feel free to use that and include the output as an image. However, please ensure that the image quality is legible and the formatting generally matches the rest of the report, and DO NOT include screenshots of raw code output—it is not professional to include dark mode or Courier New code output, or screenshots of your console window.

Table 7.1

Example text: Results for all three models over the standard benchmarking metrics. Highlighted cells show the best performance for each metric.

A table is often a better alternative to a bulleted list for communicating information that may otherwise seem redundant (e.g. variable details or model hyperparameter choices), or to make information easier to compare across categories (e.g. model performance). If you have a very complex table, think about how you might use bold or italic font to highlight notable cells, such as the best performing model.

7.3.3 Equations

Think carefully about whether including an equation is appropriate for your semi-technical stakeholder audience—most of the time that level of detail will be too technical, and equations should either be left out or included in a separate appendix.

If you decide that you do want to include an equation, they should be on a separate line, indented, and numbered sequentially to the right

of the equation. As with figures and tables, equations should be referenced directly in your text with a short description (e.g. "Equation 7.1 shows the basic linear regression equation, where y is the target variable, x is the model input, a is the learned coefficent, and b is the model intercept.") You DO NOT need to include a separate caption for your equations.

$$y = ax + b \tag{7.1}$$

As with tables, if you have a favorite tool for formatting your equations, feel free to include the formatted equation as an image, as long as it matches the formatting shown in the example above.

7.4 Works Cited

References should be alphabetically ordered using APA 7 formatting. You can refer to the APA guidelines on how to format citations from different types of sources. There are also online tools that can be used to create citations for you, though these will occasionally produce errors, so be sure to double check these. If you are citing an academic paper, the Google Scholar listing for the paper will also have a properly formatted APA citation that you can copy into your report. Several example citations are shown in the works cited included in this template.

7.5 Appendices

Appendices can be used to include information that you feel is important to the communication of your project but may be too technical or too detailed for the intended semi-technical stakeholder audience. This may include items like a detailed description of data transformations, feature engineering, or architectural choices, or complete results over a hyperparameter grid search or a series of experimental architectures.

Appendices are not included in the page count but should not be used to get around the report page limits. Information included in the appendices should not be necessary for a basic understanding of your project and should instead provide deeper detail for an interested reader. You can assume that your stakeholder audience would not read most of the content in an appendix, and as such, your appendix content may not contribute to your final report grade.

Appendices should be labeled alphabetically (Appendix A, Appendix B, etc.) and should be referred to directly in your text (e.g. "A complete table of results over all hyperparameters can be found in Appendix A"). Each appendix should start on a new page and does not need to be in the two-column format if full page width makes more sense for your content (e.g. detailed figures or large tables).

ACKNOWLEDGMENTS

These are optional, but might mention any external contributors to your project, such as professors or mentors who provided data or consultation.

Works Cited

American Psychological Association. (2025). *In-Text Citations*. https://apastyle.apa.org/style-grammar-guidelines/citations

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171-4186).

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI blog, 1*(8), 9.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems, 30.*